Generalization and metric divergence in overparameterized Conformer-CTC ASR

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Project idea

Classical theory: test error rises once model capacity exceeds data complexity – more parameters means overfitting.

Modern practice: very large models often keep improving on held-out data well past the point where training loss reaches zero.

This work asks two questions in the context of speech recognition:

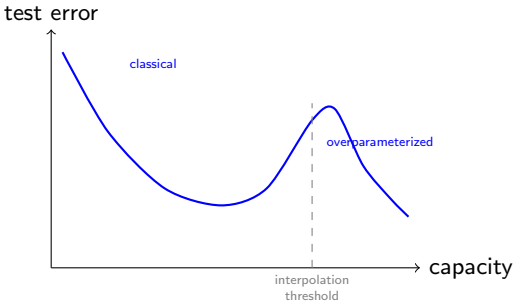
1. Does increasing Conformer capacity in ASR trained on CTC (Connectionist Temporal Classification) loss produce a double-descent pattern in generalization?
2. Do CTC loss and word error rate agree on what "generalization" means, or can they diverge in the overparameterized regime?

This is the first empirical study of double descent in a CTC-based sequence recognition system.

Theory background

Double-descent risk curve (Belkin et al. 2019; Nakkiran et al. 2019)

Past the interpolation threshold, test error can resume decreasing. Model-wise, sample-wise, and epoch-wise variants documented. Parameter count alone does not explain generalization (Zhang et al. 2021).



CTC loss (Graves et al. 2006)

- Marginalizes over all monotone input–output alignments
- Low train loss $\neq$ low WER: good alignments found, but decoded sequence may still be wrong

Conformer (Gulati et al. 2020)

- Self-attention + depthwise convolution per block
- FF $\rightarrow$ Attn $\rightarrow$ Conv $\rightarrow$ FF, pre-norm residuals
- 5 LayerNorm + 1 BatchNorm per block

Implementation

Model

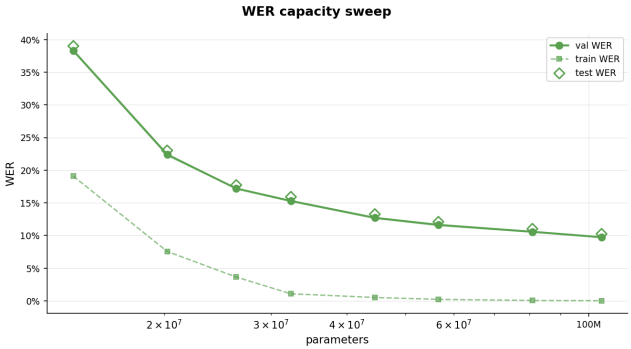
- Conformer-CTC, depth sweep $L \in \{1, 2, 3, 4, 6, 8, 12, 16\}$
- Fixed hidden dim $d = 512$, 8 attention heads
- 14 M parameters at $L = 1$ to 105 M at $L = 16$
- Interpolation criterion: train loss below 0.01
- Current sweep capped at 105 M – extending further is planned for Stage B

Training

- LibriSpeech train-clean-360+100 ($\approx$460 h)
- 100 000 optimizer steps, batch 16
- Adam + warm-up cosine LR, 2 seeds per config
- Test-clean held out until final evaluation
- Subset ablation: 5k / 50k / 150k utterances with the same budget

Logged per checkpoint: train/validation/test loss, WER (greedy decoding), generalization gap, weight norm, sharpness proxy.

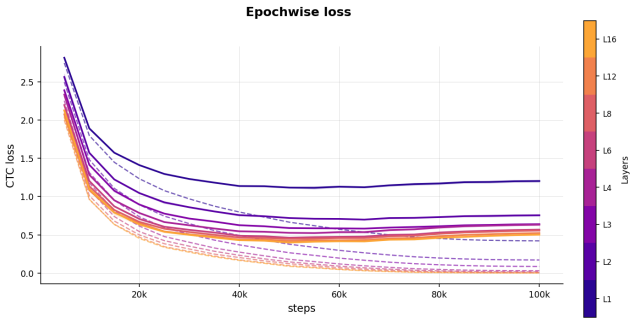
Result 1: WER decreases monotonically with depth



L	Params	Test WER
1	14 M	39.0%
2	20 M	23.0%
4	32 M	15.9%
8	57 M	12.1%
12	81 M	11.0%
16	105 M	10.3%

No double-descent peak in WER. Larger models generalize better under clean labels within the training budget.

Result 2: validation loss and WER diverge past interpolation



Solid: val loss. Dashed: train loss.

Models with 57 M parameters and more reach the interpolation threshold by the end of training (train loss below 0.01).

After interpolation:

- Validation loss rises slightly – classical overfitting signal
- Validation WER keeps improving

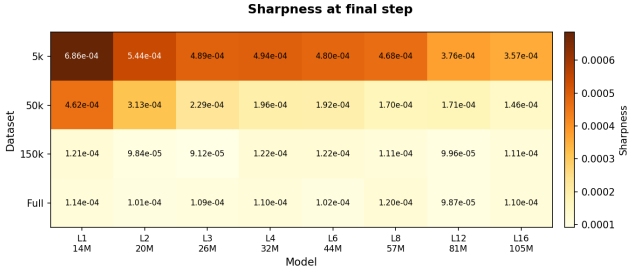
Greedy decoding can improve even when cross-entropy rises. CTC loss is not a reliable proxy for recognition quality past interpolation.

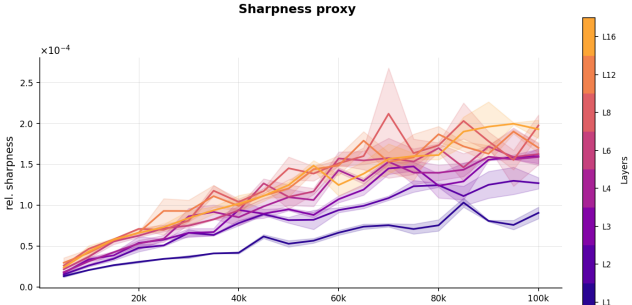
Result 3: subset ablation

Test WER at the end of training across data scales and model depths.

- Small subsets (5k, 50k): all models reach interpolation. Larger models still get lower WER – more capacity leads to better generalization even when every model memorizes the training set.
- Large subsets (150k, 460k): small models underfit, large ones approach interpolation.
- No double-descent peak in any tested regime.

Sharpness proxy





One-sided random-perturbation estimate (absolute $\widehat{S}$, top; relative $\widehat{S}/\mathcal{L}_{\text{test}}$, bottom):

$$\widehat{S} = \mathbb{E}_\delta[\max(0,\ \mathcal{L}(\theta+\delta) - \mathcal{L}(\theta))]$$

$$\delta = \frac{\varepsilon_{\text{rel}}\,\|\theta\|_2}{\|\xi\|_2}\,\xi, \quad \xi \sim \mathcal{N}(0, I), \quad \varepsilon_{\text{rel}} = 10^{-3}$$

Five LayerNorm and one BatchNorm per block absorb weight perturbations before they reach the output. Values are uniformly small ($\approx 10^{-4}$) with no consistent ordering by depth – the proxy cannot be treated as a reliable curvature estimate for this architecture.

Conclusions

1. No double-descent peak in WER under clean labels on 460 h of speech within the training budget.
2. Validation loss and WER diverge past interpolation: validation loss rises slightly while WER keeps improving. CTC loss overstates generalization failure in the overparameterized regime.
3. More capacity leads to better generalization in every tested regime – both when models underfit and when they fully memorize the training set.

Future work – Stage B

To observe the double-descent peak

- Add controlled label noise – theory predicts a sharper generalization peak near the interpolation threshold
- Run subset experiments to convergence (fixed epochs, not fixed steps) to decouple data scale from optimization time
- Extend the capacity sweep beyond 105 M parameters

Generalization theory for CTC

- Build a theoretical framework to bound generalization under CTC loss – starting from Rademacher complexity and accounting for the latent alignment structure of CTC

Better sharpness measures

- Replace the current perturbation proxy with SAM-based sharpness: SAM (Sharpness-Aware Minimization, Foret et al. 2021) finds the worst-case loss within a parameter neighborhood by solving a small inner maximization at each step – the perturbation radius gives a principled, geometry-aware curvature estimate that is robust to normalization layers
- Or use filter normalization to remove the scale ambiguity introduced by LayerNorm before computing sharpness

References (1/3)

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